

Additional visuals not already covered

2026-06-18

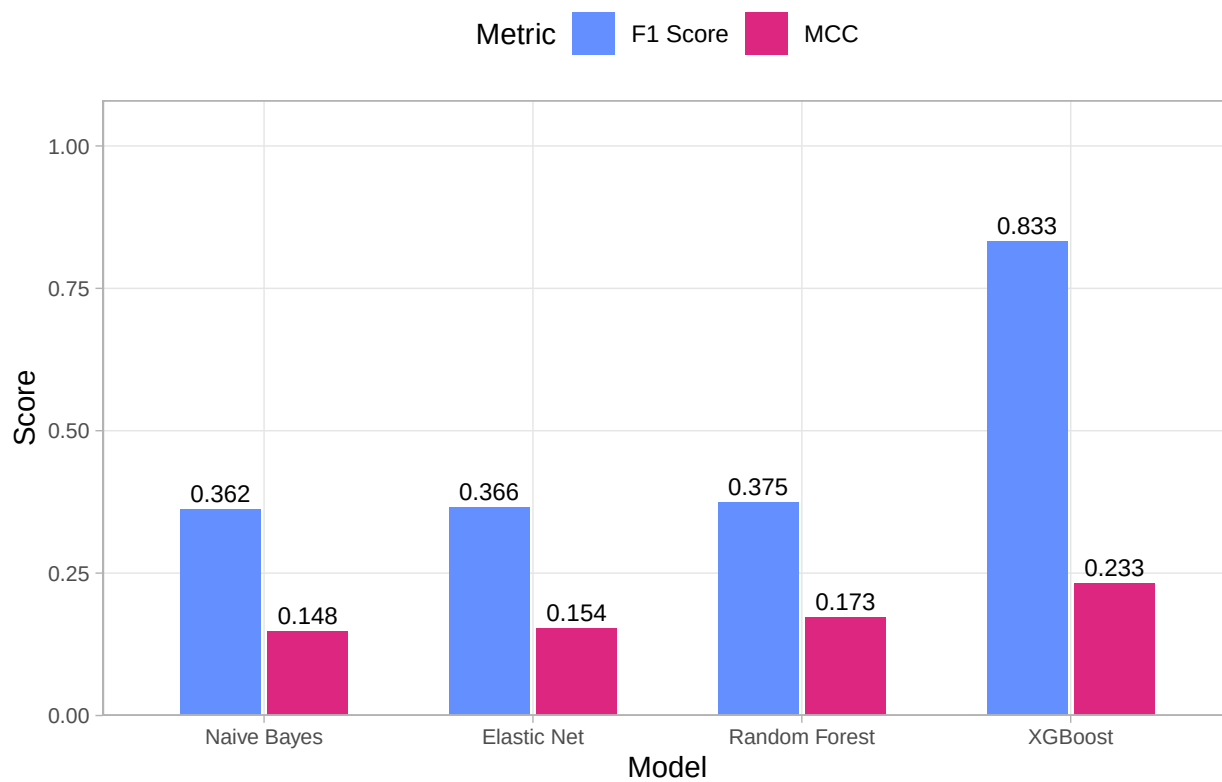
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1 Cross Model Comparisons

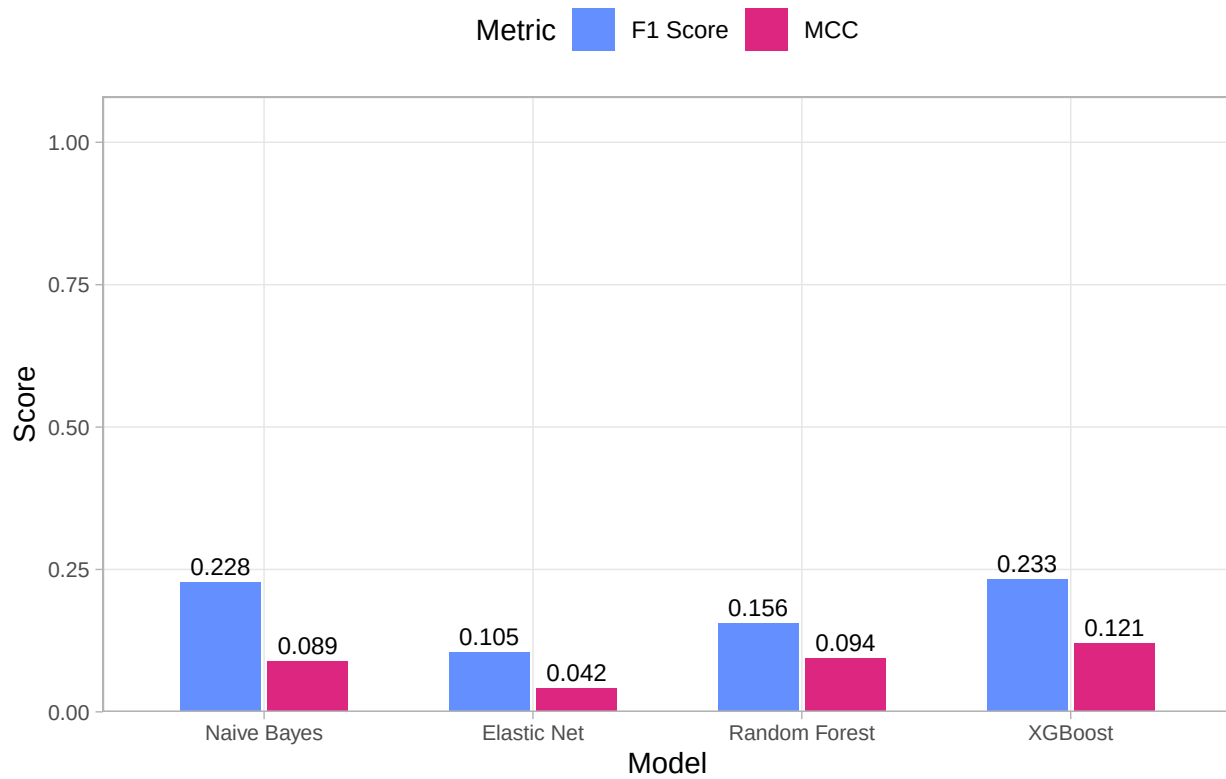
1.1 Crime Type

Crime Type Model Performance: F1 Score and MCC



1.2 Crime Subtype

Crime Subtype Model Performance: F1 Score and MCC



2 Naive Bayes

2.1 Equations

$$P(\text{Violent} \mid \mathbf{x}) \propto P(\text{Violent}) \prod_{j \in S} P(x_j \mid \text{Violent})$$

3 Elastic Net

3.1 Tables

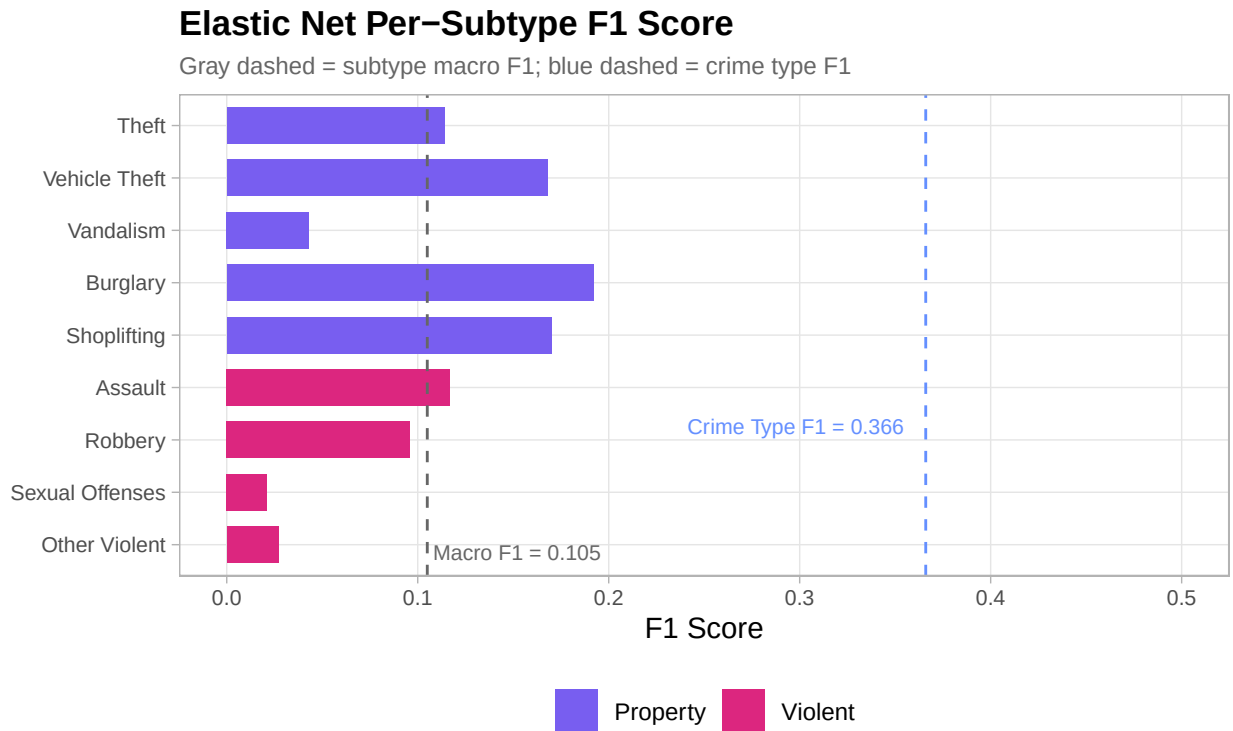
Table 1: Elastic Net: Crime Type Classification Metrics

Metric	Value
Primary Metrics	
F1 Score	0.366
MCC	0.154
Overall	
Accuracy	0.560
AUC	0.627
Sensitivity	0.659
Specificity	0.536
Precision	0.253
Balanced Accuracy	0.598

Table 2: Non-Zero Elastic Net Coefficients

Predictor	Coefficient
College Degree Rate	-0.820
Single Parent Rate	0.713
No Vehicle Rate	0.629
No HS Diploma Rate	0.546
Disability Rate	0.208
Cluster: Low Disadvantage	-0.101
Internet Access Rate	-0.092
Poverty Rate	0.065
Cluster: Highest Disadvantage	0.058

3.2 F1 Graph



3.3 Equations

$$L(\beta) = -\frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{p}_i) + (1 - y_i) \log(1 - \hat{p}_i)] + \lambda \left[\alpha \sum_{j=1}^p |\beta_j| + \frac{1 - \alpha}{2} \sum_{j=1}^p \beta_j^2 \right]$$

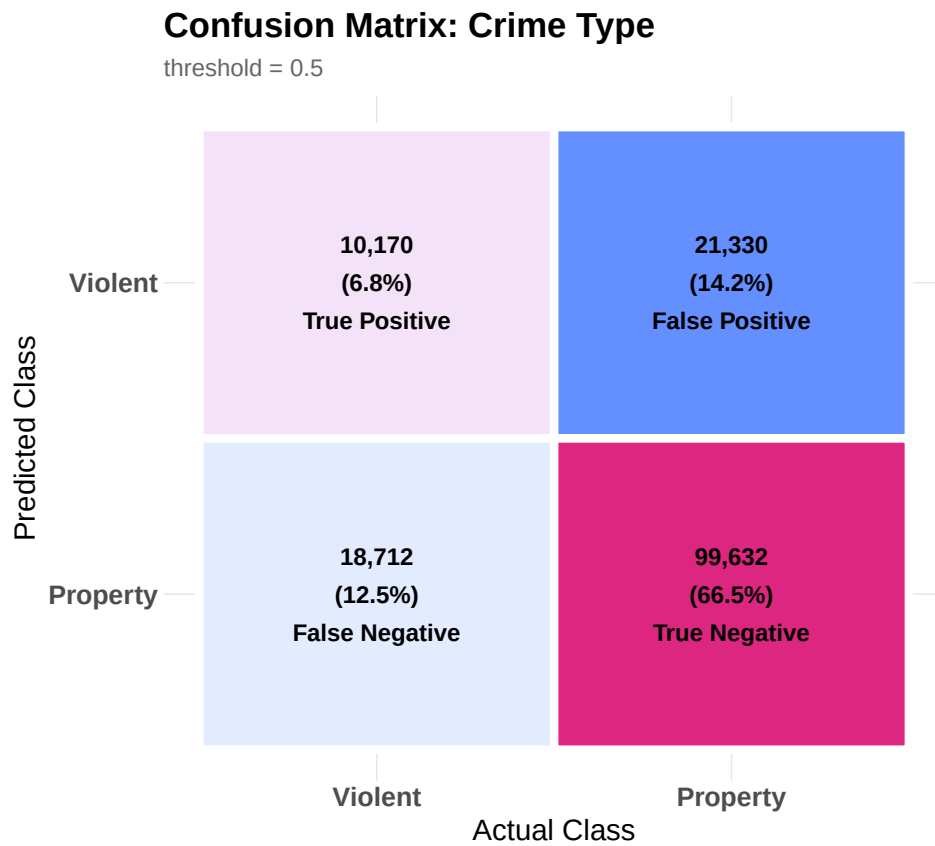
4 Random Forest

4.1 Equations

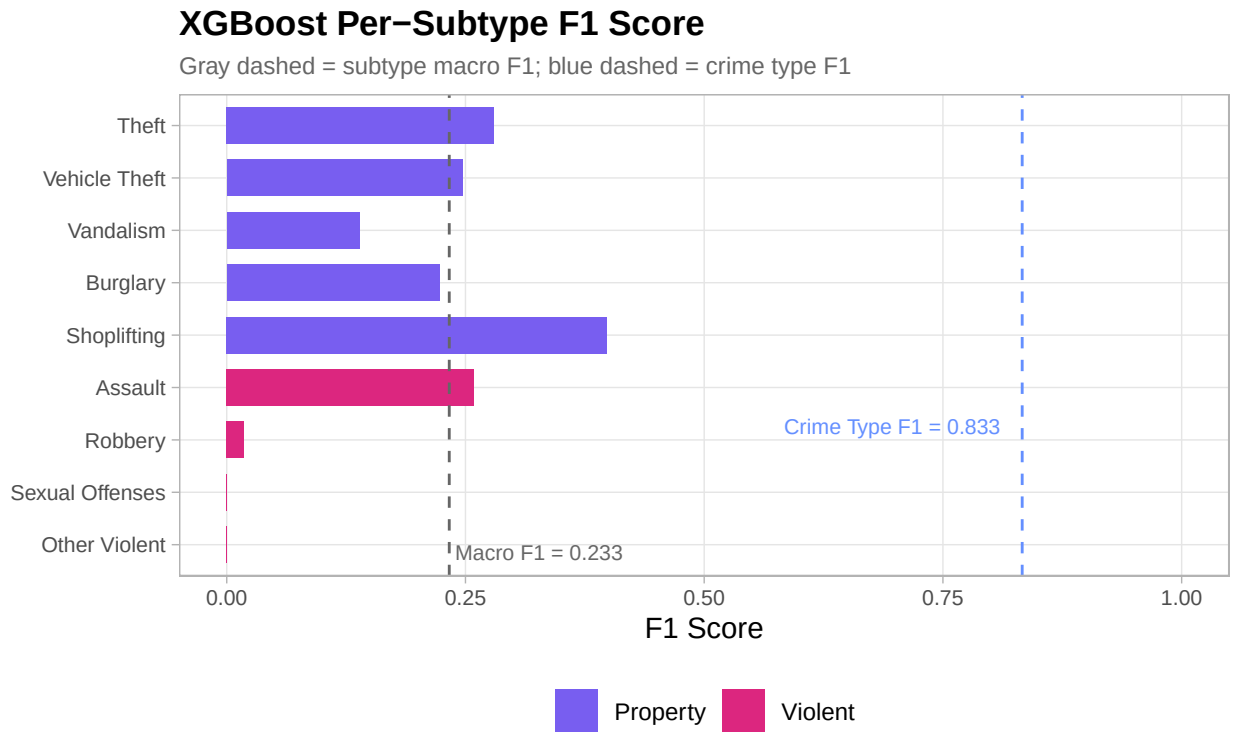
$$\hat{y}_i = \mathbb{1} \left(\frac{1}{B} \sum_{b=1}^B T_b(\mathbf{x}_i) > \tau \right)$$

5 XGBoost

5.1 Confusion Matrix



5.2 F1 Graph



6 RQ2

6.1 Equations

$$y_i = \beta_0 + \sum_{k=1}^p \beta_k x_{ik} + \varepsilon_i$$

$$\hat{y}_i = 1.724 - 0.159(\text{Median Income}) + 0.235(\text{No Vehicle Rate}) + 0.088(\text{No HS Diploma Rate}) + 0.053(\text{Single Parent Rate})$$

$$y_i = \beta_0(u_i, v_i) + \sum_{k=1}^p \beta_k(u_i, v_i) x_{ik} + \varepsilon_i$$

$$\hat{\beta}(u_i, v_i) = (\mathbf{X}'\mathbf{W}(u_i, v_i)\mathbf{X})^{-1} \mathbf{X}'\mathbf{W}(u_i, v_i)\mathbf{y}$$

$$y_i = \beta_0(u_i, v_i) + \sum_{k=1}^p \beta_{h_k, k}(u_i, v_i) x_{ik} + \varepsilon_i$$